

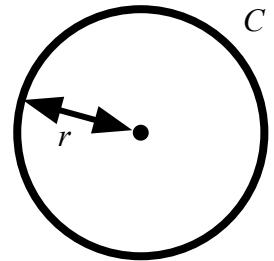


Changing the subject to suit the context

Task A: Rearranging additive and multiplicative steps

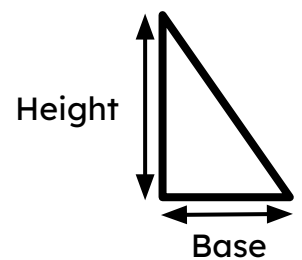
1) a) Rearrange the formula for circumference of a circle so that it would be more efficient for finding the radius if you knew the circumference.

$$C = 2\pi r$$



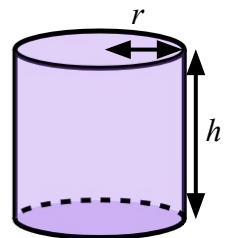
1) b) Rearrange the formula for area of a triangle so that it would be more efficient for finding the height if you knew the area and the base.

$$A = \frac{1}{2}bh$$



1) c) Rearrange the formula for surface area of a cylinder so that it would be more efficient for finding the height if you knew the surface area and radius.

$$A = 2\pi r^2 + 2\pi rh$$



2) This is the formula for converting Celsius into Fahrenheit.

Show two ways to rearrange the formula so that it converts Fahrenheit into Celsius.

$$F = \frac{9c}{5} + 32$$

Task B: Rearranging equations to find the gradient

1) Work out the gradient and the y -intercept of the lines given by these equations by rearranging to make y the subject.

a) $x + 4y = 12$

b) $2x + 6y = -18$

c) $8x - 2y = 15$

2) Rearrange this general form of a linear equation to make y the subject and find an expression for the gradient and y -intercept when in this form.

$$ax + by = c$$